

# Replication Report: Quantitative Relationship between Polarization Differences and the Zone-Averaged Shift Photocurrent

OSTI 1523841 | Coverage 8/10, Agreement 10/10

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## 1 Source paper

- **Title:** Quantitative Relationship between Polarization Differences and the Zone-Averaged Shift Photocurrent
- **Authors:** Fregoso, Morimoto, Moore
- **Year:** 2017
- **Domain:** Condensed Matter / Topological Response / Shift Current Photovoltaics
- **Identifier:** OSTI 1523841

## 2 Replication objective

The central claim of the paper concerns condensed matter / topological response / shift current photovoltaics. Our objective was to reproduce the headline numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72 h, commodity GPU/CPU compute, no author contact). A replication plan is archived at [1523841-Quantitative-relationship-between-polarization-differences-and-the-zone-averaged-shift-photocurrent](#).

## 3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper’s computational tools inventory (see [COMPUTATIONAL\\_TOOLS\\_INVENTORY.md](#)).

### Paper-directory README excerpt

**Quantitative relationship between polarization differences and the zone-averaged shift photocurrent**

- **\*\*OSTI ID:\*\*** 1523841
- **\*\*Rank:\*\*** #22
- **\*\*Replication Score:\*\*** 9/10
- **\*\*Open-Source Tools:\*\*** Yes

- **Code Repository:** No
- **Tools:** None explicitly (tight-binding models, analytic equations; implementation possible in Python, Julia, etc.)

### # Why This Paper

Purely analytic/numerical, all equations and parameters specified, synthetic data, and quantitative results. No code repo but trivial to implement for AI.

### # Replication Plan

AI implements tight-binding models, computes Berry connections/shift vectors, and reproduces tables/curves using open-source tools.

### # Status

- [ ] Paper reviewed
- [ ] Code/tools identified
- [ ] Code implemented/cloned
- [ ] Results reproduced
- [ ] Results validated against paper

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper’s resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

## 4 What was reproduced

- All previous items (Eq.9, Eq.C5, Eq.D4, Eq.D8-D9, winding discontinuity, Fig.1(b))
- Multi-band Wilson-loop/Eq.17 verification on 3-band trimer
- Multi-band test on 4-band coupled RM model
- 1D BHZ-like spin-doubled model ( $W=0$  trivial check)
- Integer winding  $W_{cv}$  for every band pair (gauge-independent)
- Berry polarizations  $P_n \bmod 1$  invariance under parallel transport

## 5 What was missing or incomplete

- Material-specific DFT+Wannier (GeS, BaTiO<sub>3</sub>, monolayer WS<sub>2</sub>)
- Frequency-integrated shift-conductivity connection  $\hat{\sigma}^{\text{shift}}(\omega)$
- Paper Figs 2-4 (higher-D extensions, material examples)
- Comparison vs experimental shift-current measurements
- Hartree-Fock/GW interacting-system extension

The dominant blocker categories for this paper, mapped onto the meta-paper’s 9-category friction taxonomy (§4), are listed in §?? below.

## 6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

**Agreement rationale.** Two-band identity continues to hold to machine precision on the Rice-Mele model. New multi-band tests confirm gauge-invariant polarizations and integer winding numbers (BHZ:  $W=0$  trivially; trimer/RM-block: integer winding tracked through level reordering). Identity  $(P_c - P_v) + W_{cv}$  is preserved as the universal structure across all tested band pairs.

## 7 Score and friction-point tagging

- **Coverage:** 8/10 — Original  $c=7$  lifted to 8 by adding a multi-band tier-lift extension (code/multiband\_extension.py) that verifies the polarization-difference identity beyond the two-band Rice-Mele case. We tested the Wilson-loop/Eq.17 generalisation on (1) a 3-band trimer chain, (2) a 4-band coupled Rice-Mele block model, and (3) a 1D BHZ-like spin-doubled model. Berry polarizations are gauge-invariant mod 1, the dipole-phase winding  $W_{cv}$  is integer for every band pair, and the residual  $P_c - P_v + W_{cv}$  is well-defined. This addresses follow-on Q1 directly. Still missing: material-specific DFT+Wannier calculations (GeS, BaTiO3) and the frequency-integrated shift-conductivity connection (Q2/Q3).
- **Agreement:** 10/10 — Two-band identity continues to hold to machine precision on the Rice-Mele model. New multi-band tests confirm gauge-invariant polarizations and integer winding numbers (BHZ:  $W=0$  trivially; trimer/RM-block: integer winding tracked through level reordering). Identity  $(P_c - P_v) + W_{cv}$  is preserved as the universal structure across all tested band pairs.

**Friction points encountered (taxonomy tags):**

- **F7:** Missing surrogate calculators (xTB/DFT/closed solver)

## 8 Follow-on questions

- Q2/Q3/Q5 from prior entry remain open

## 9 Reproducibility pointers

- **Paper directory:** 1523841-Quantitative-relationship-between-polarization-differences-a
- **Replication artefacts:**
  - /Users/stevens/Dropbox/REPLICATE-PROJECT/1523841-Quantitative-relationship-between-po
- **Replication-dir hint from JSONL:** ~/Dropbox/REPLICATE-PROJECT/1523841-Quantitative-relations
- **Status:** tier\_lift\_2026\_04\_26

To re-run, change directory into the paper directory above and consult its README.md for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.